

DAA Practical 2

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#include <iostream>
#include <vector>
#include <chrono>

using namespace std;
using namespace std::chrono;

// Linear Search Function
int linearSearch(vector<int> arr, int key)
{
    for (int i = 0; i < arr.size(); i++)
    {
        if (arr[i] == key)
            return i;
    }
    return -1;
}

// Binary Search Function
int binarySearch(vector<int> arr, int key)
{
    int low = 0;
    int high = arr.size() - 1;

    while (low <= high)
    {
        int mid = low + (high - low) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }

    return -1;
}

int main()
{
    int n = 100000;
    vector<int> arr(n);

    // Creating a sorted array
    for (int i = 0; i < n; i++)
        arr[i] = i + 1;

    int key;
    cout << "Enter element to search: ";
    cin >> key;

    int index;

    // Linear Search Timing
```

```

auto start = high_resolution_clock::now();
index = linearSearch(arr, key);
auto stop = high_resolution_clock::now();

cout << "\nLinear Search\n";
if (index != -1)
cout << "Element found at index " << index << endl;
else
cout << "Element not found\n";

cout << "Time Taken : "
<< duration_cast<microseconds>(stop - start).count()
<< " microseconds\n";

// Binary Search Timing
start = high_resolution_clock::now();
index = binarySearch(arr, key);
stop = high_resolution_clock::now();

cout << "\nBinary Search\n";
if (index != -1)
cout << "Element found at index " << index << endl;
else
cout << "Element not found\n";

cout << "Time Taken : "
<< duration_cast<microseconds>(stop - start).count()
<< " microseconds\n";

return 0;
}

```